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# Optimization-Based Traffic Safety Improvement Strategy for Autonomous Vehicle Driving at Level Crossings

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**Abstract**

Railway level crossings pose a serious threat to the safety and mobility of drivers traversing them. Globally, a significant number of traffic violations occur annually at level crossings. This study proposes an optimization-based autonomous vehicle (AV) driving strategy to improve the traffic safety of AVs at level crossings. First, an optimization model considering the operational constraints is developed. The cost function of the proposed optimization-based driving strategy is the time difference between the time required by the train to reach the level crossing and the time taken by the AV to complete the whole crossing maneuver. Second, a MATLAB simulation model is developed to validate the proposed driving strategy. Simulation results show that the proposed driving strategy delivers optimized AV driving without stoppage. The comparative study also shows that the proposed method improves driving performance. For instance, when a train is 1 m away from the light signal while the AV's initial speed is 36 km/h, the time required for the AV to reach the speed limit sign is about 0.44 s, while the time required by the train to reach the car block area is 0.13 s. Thus, the AV is forced to stop and then go after the train leaves the level crossing. Contrary to this, the proposed method allows a safe passage of the vehicle within 1.66 s without stoppage. Most importantly, the driving strategy is a collision avoidance strategy without unnecessary enforcement of stop-and-go when AVs have the chance of crossing safely. Consequently, the optimization-based driving strategy both improves traffic safety and reduces traveling time.

**Keywords**

safety, transportation safety management systems, rural road safety, safe systems, vehicle technologies

Level crossings (also known as grade crossings) present one of the highest risk factors for motor vehicle accidents, which are often caused by the unpredictable behavior of drivers. Traffic signals operate at level crossings to prevent collisions between motor vehicles and trains. It is nearly impossible for a train to come to a complete stop in time to avoid colliding with a vehicle that has stopped on a level crossing. The vehicle's presence by itself is a threat to safety, whether there is a collision or not. In that regard, the implementation of intelligent modes of transportation is necessary under such circumstances. It improves road safety and comfort by reducing the causes of traffic collisions, for instance, using information and communication technologies to reduce traffic violations. Among these technologies, vehicle-to-infrastructure and vehicle-to-vehicle communications are promising candidates for improving traffic safety performance (*1*).

Furthermore, traffic safety has become a global concern as road traffic presents a risk to human lives and property. Statistical reports on road traffic violations indicate that in the U.S.A. 8% of total annual violations

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occurred at freeway intersections in 2019, with a 13% mortality rate (*2*). The cost of fatalities, injuries, and property damage was estimated to reach $413.8 billion globally in 2017. Annually, up to 50 million people worldwide suffer nonfatal injuries as a result of traffic crashes (*3*). Moreover, some previous studies contended that 90% of traffic violations occur because of the failure of drivers when multiple tasks are performed in parallel (*4*, *5*). Nallet et al. (*6*) and Tseng (*7*) further affirmed that traffic rules are violated by more than 50% of drivers of conventional vehicles in France. That implies the necessity of major investigations to improve road safety and comfort by identifying high risk factors besides incorporating additional traffic safety procedures.

To that effect, one of the major strategies is to use traffic signs as the first line of warning and as a safety protection procedure to control possible violations at intersections. For instance, a "Stop Ahead" or "Yield" sign procedure has been used to improve road traffic safety in a conventional driving strategy (*8*). However, because of various risk factors, this safety procedure fails adequately to inform drivers of the actual presence of trains at level crossings. According to statistics, serious injuries and deaths are major risk factors for pedestrians and vehicles. In this regard, 27% of traffic accidents in Florida resulted in serious injury and death, and 19% in the U.S. nationwide (*9*). Several empirical studies also illustrate that conventional drivers violate traffic signals because of driver reaction time and underestimation of traffic signals (*10–12*).

Previous studies have found that drivers' behavior in mixed traffic significantly affects traffic safety (*13*). Moreover, driver behavior may trigger traffic violations, and violations, in turn, cause traffic collisions. In addition, the effects of lead vehicle size, traffic composition and flow, and headway distributions on longitudinal vehicle behaviors could be additional risk factors for traffic violations (*14*). A recent study on road traffic safety as a proxy identified high levels of traffic violations in Luzhu, China, both in violation of traffic signs and illegal parking violations (*15*). Cheng et al. (*16*) and Cooper (*17*) reported that speeding and underestimation of critical safety distances are serious and common issues globally. Most importantly, it was found that 27% of drivers with one or more prior records of violation tended to violate level crossings (*18*). The same study further indicated that drivers tended to violate level crossing signals based on their risk assessment, which was influenced by the actual sight of the train, judgment of the time before the train will arrive at the crossing, and perceived speed of the train, respectively.

Under normal conditions, for an approaching vehicle to pass through the level crossing without violation, the signal at the level crossing must display a green light. On the other hand, when a train approaches the crossing and receives this information from the interlocking system, the road signal system turns on the red light after flickering the green light for 5 s (*19*). Accordingly, it compels vehicles and pedestrians not to enter or speed up to pass the level crossing. The system ensures that when the crossing is clear, the road signal automatically recovers to green and continues cycling. Moreover, constant warning time (CWT) systems are installed to improve safety by detecting the train as it approaches, calculating its speed and distance, and activating the traffic control devices to provide the desired warning time (*20*). However, this only applies to the conventional stop-and-go driving strategy (*21*). Benchmarking the findings of Khoudour et al. (*22*), which attest to violation of stop signs by 14% of drivers and 29% of cases related to level crossing operation errors, it is of paramount significance to espouse other strategies for reducing collisions at level crossings. A wealth of research supports the potential for autonomous vehicles (AVs) to maintain critical safety while driving (*23*).

AV is regarded as one of the future safer driving strategies capable of easily detecting obstacles in time to avoid risky conditions (*24–26*). Studies in this line have also shown significantly reduced collisions, particularly at intersections (*27–29*). For instance, with obstacles found at an average distance of 0.321 m, AVs can automatically stop, accelerate, or decelerate (*30*, *31*). However, recent developments hint at the sharing of existing road facilities by conventional vehicles and AVs. To the contrary, conventional vehicles, with stop-and-go driving strategy, are required to stop at speed limit signs between 4.5 and 15 m from the level crossing (*32*). In this regard, AVs may contribute significantly to avoiding the stop-and-go driving strategy, as well as revolutionizing mixed traffic safety standards.

Collisions at level crossings are likely to continue to occur as a result of infringements by drivers of conventional vehicles, heterogeneity of vehicles at level crossings, and other factors. Therefore, this study provides an optimization-based traffic safety improvement strategy for AVs passing level crossings at the speed limit without stopping, to avoid traffic losses and to reduce traveling time even if the train enters the light signal region. To that end, this study adopts a MATLAB simulation approach to validate the proposed optimization-based AV driving strategy. Overall, attempts are made to consider numerous operating conditions and different influence factors.

The study makes the following major contributions. First, an optimization model is developed where the timed difference between the journey time of the train to the level crossing and the total time taken by the AV to clear the level crossing is the objective function. Second,

a MATLAB simulation model is developed to validate the optimization model. Third, a comparative study with the stop-and-go approach is given, where the proposed optimization-based driving strategy adds considerable value to the existing literature and current knowledge. Finally, while considering the role of AVs in a mixed traffic environment, the study suggests possible future policy implications. Consequently, the findings will be beneficial for conceiving safety treatment strategies against traffic violations and stop-and-go, and used as inputs for the future direction of safe driving strategy.

The remaining sections of this study are organized as follows. The second section presents a review of related studies. The third section describes the proposed method and procedures in detail. The fourth section describes the simulation performed to evaluate the performance and discusses the results. The final section provides the conclusions of the study.

## Related Work

Level crossing safety management is a shared responsibility between rail and road operators. However, traffic violations are expected due to risky driving strategies. The majority of violations of level crossings are caused by driver negligence, and they usually result in significant human injury, fatality, and material damage to the vehicles involved as well as the safety of trains (*16, 33–35*). Thus, to counter the possibility of collision, drivers are expected to abide by the traffic rules such as driving slowly, as per the posted speed limit sign, and yielding or stopping as required before entering the level crossing.

Previous studies have proposed several solutions to address traffic violation issues associated with intersection crossing maneuvers. Centralized intersection crossing coordination, discretization of space, and the impact factors of collision using a hybrid dataset were the most relevant to the analysis of the longitudinal aspect of the maneuver (*36*, *37*). Lenné et al. (*38*) found that coordinating passive and active traffic crossing signs alongside intersections was more important for reducing traffic queues and improving traffic flow near the crossings. Apart from violations, other contextual factors like traffic light status, driver perception, and distance to the roadside may significantly affect conventional driving strategy. Consequently, it can be seen that the stop-and-go driving strategy significantly affects the implementation of mixed traffic and, in turn, the merits of avoiding traffic violations and the resulting safety improvements (*10–12*).

Stop signs at level crossings help maintain orderly traffic, allow pedestrians or vehicles to cross, and reduce collisions between vehicles and trains entering the crossing. The conventional driving mode enforces safety procedures that may involve vehicles reaching a complete stop before entering an intersection crossing. Read et al. (*39*) presented relationships between the identified factors like risk and magnitude, unsafe and unsatisfactory road user characteristics, and road user risk perception, attitudes, and beliefs. It indicates that the main consideration has been given to the intersection of vital factors (i.e., how crossing design influences driving near the roadside). A study was also conducted to determine why drivers enter level crossings when trains are crossing (*40*). The study results indicated that drivers mostly decide to enter or not stop safely before crossing the railway, which is related to the appropriate and stressful zones, and it is important to stop carefully before entering level crossings. Other studies have been conducted to assess the behavior and safety of drivers at high-speed train crossings (*41*, *42*). These studies found that most drivers who cross the tracks at the warning pass within 5 s of reaching the crossing. However, most drivers expect the train to arrive at the railroad crossing within 20 s. It was concluded that drivers would lose confidence in the traffic control system if the warning time were more than 40 s with flashing lights and 60 s with closed interruptions. However, the role of AVs was not considered.

Furthermore, a study has focused on analyzing driver behavior in relation to the various phases of the interlocking system (*19*). On the other hand, qualitative methods were employed to understand the driver's behavior throughout the entire cycle of intersection controls, including when the road is open to traffic (*16, 33–35*). Moreover, other studies employed qualitative and quantitative methods based on surveys, interviews, and focus groups to identify the influence factors (*43–45*). Young et al. (*46*) and Rudin-Brown et al. (*47*) examined where drivers direct their attention while approaching the intersection and whether this differs between novice and experienced drivers. These studies, using a driving simulator, also compared the efficacy and drivers' subjective perception of level crossing traffic control devices such as flashing lights with boom barriers and standard traffic lights. Accordingly, they found that useful insights into the approaches to urban level crossings with regard to drivers' visual and cognitive behaviors and traffic lights at real-world crossings may not provide safety benefits, respectively.

In the same vein, Venglar (*46*) proposed an "Improved Transition Preparation Strategy" (ITPS) to enhance mobility and safety near crossings. This investigation added more logic tests to the ITPS algorithm, aiming to provide more green time for the conflicting phases with a train crossing. The results indicate the ITPS improved average intersection crossing delay. In the study by Cho and Rilett (*48*), the TPS algorithm operated until the normal CWT system was activated and signal control was set in advance. This reduced the

efficiency of long cleaning steps and crossings before the train entered the CWT detection cycle. Since these dwell phases are served again during train crossings, the TPS may provide inordinate green time to those cycles, resulting in longer average intersection delays.

Furthermore, previous studies found that conventional vehicle drivers' reaction times and eye heights were 0.8–1.5 s and 1,080 mm, respectively (*49–51*). However, with the improvement of these two values in level-5 driverless AVs to 0.001–0.01 s and 360° surrounding view cameras on each side of the vehicle (*52*, *53*), it seems reasonable to suggest that AVs offer improvement in level crossing collisions against all the odds. Although previous studies have confirmed the superiority of AVs compared with conventional vehicles (*49–51*), there are still gaps in considering the role of AVs in avoiding traffic jams and traffic safety procedures to improve traffic safety at level crossings. Overall, given the limitations of stop-and-go driving strategy in heterogeneous traffic environments and the promises of AVs, it is of paramount importance to research optimization-based level crossing operation approaches.

## Research Methodology

This study demonstrates an optimization-based driving strategy for determining the maximum time difference between an AV and a train (this paper particularly addresses light-rail transit [LRT]). The conventional stop-and-go driving principles were simulated using the positions of AV and LRT to illustrate the effects of the conventional driving strategy at crossings. Moreover, establishing an optimization-based driving algorithm ensures safe AV driving without stopping at speed limit signs. LRT typically operates through level crossings in semi-exclusive and mixed-use alignments with speed limits ranging from 16.1 to 104.5 km/h (*32*). It is assumed that the LRT leaves the crossing at a constant speed of 28.8 km/h after entering the car block area (CarBlockArea). The optimal critical safety distance and speed limit can thus provide a safe trajectory for an AV while passing through the crossing.

### *Conventional AV Stop-and-Go Driving Strategy Model*

A stop-and-go driving strategy is a regular practice to ensure traffic safety and minimize driver errors at crossings. This means that human drivers have to drive cautiously. For instance, they might slow down, prepare to stop, look both ways cautiously, and not enter the level crossing unless it is possible to drive completely through without stopping. Some drivers look again before deciding to cross and keep going once they start. To mitigate the risk, safety countermeasures for both active and passive crossings have been implemented (*54*). However, quantitative studies indicate that traffic volume is increasing, and a high likelihood of collision occurrence involving violators is predictable (*55*, *56*). In addition, studies have found that some of the common issues, particularly at crossings, include without any warning signal or protection equipment and visibility that might not allow drivers to see an approaching train (*55*). Consequently, the conventional stopping driving strategy might be a vital risk factor in mixed traffic safety. In general, the following subsections indicate a typical stop-and-go driving strategy.

*When LRT Is Inside the AV Block Region.* We assumed that in the stop-and-go model two driving states are possible. First, when the LRT is inside the CarBlockArea, by definition, the AV must stop at a safe distance.

$$S_{AV} = V_{AV} T_{\tau_{AV}} + \frac{V_{AV}^2}{2b} + L \tag{1}$$

$$T_{AV} = T_{LRT} - T_{DnAV} = \frac{S_{LRT}}{V_{LRT}} - \frac{V_{AV}}{b} - T_{\tau AV},$$
$$S_{LRT} = T_{rCl} + R_{dW} + T_L + Y \tag{2}$$

where $S_{AV}$ is the distance traversed by the AV before reaching the car speed limit (CarSpeedLimit). This can be obtained from the current speed of the AV. $V_{AV}$ is the AV's initial speed, $T_{\tau AV}$ is the AV's reaction time, $L$ is the AV's length, and $b$ is the AV's deceleration rate, as expressed in Equation 1. $T_{AV}$ is the total time of the AV traversing to complete the whole journey from its current location to clear the level crossing, and $T_{LRT}$ is the traversing time of the LRT to complete the journey from its current location to the level crossing. $T_{DnAV}$ is the acceleration and deceleration time of the AV, $S_{LRT}$ is the total distance of the LRT from the current location to clear the crossing. Likewise, $V_{LRT}$ is the LRT's initial speed, $T_{rCl}$ is the LRT's clearance distance, $R_{dw}$ is the width of the level crossing, $T_L$ is the LRT's length, and $Y$ is the distance to the LRT's location when inside the CarBlockArea or approaching the light signal, as expressed in Equation 2. Similarly, the LRT distance is straightforward ($S_{LRT}$). Therefore, after reaching the CarSpeedLimit ($S_{AV} = 0$), the AV should stop for a minimum time of $T_{AV}$ during the LRT traversing period, as shown in Figure 1.

*When LRT Is Outside the AV Block Region.* This operating condition is when the LRT either approaches the light signal or clears the level crossing. In this condition, the regular operation of the AV is possible. Depending on the current speed of the AV, the AV accelerates or decelerates until it reaches the CarSpeedLimit, where a level crossing speed limit is asserted. If the LRT is

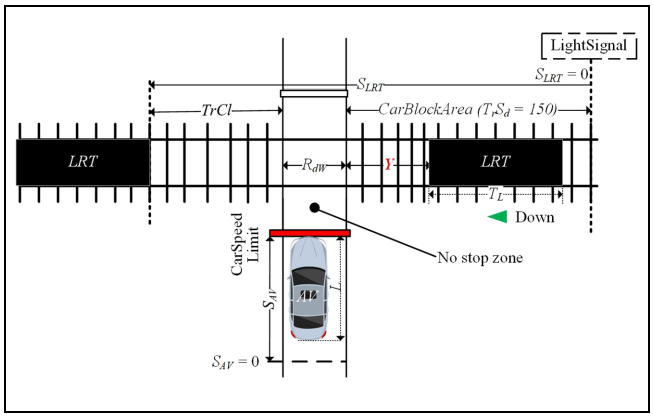



**Figure 1.** Positional relationship when train (light rail transit [LRT]) is inside CarBlockArea.

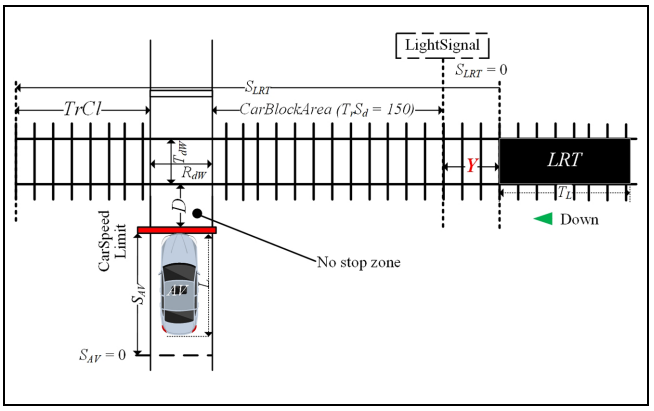



**Figure 2.** Positional relationship when train (light rail transit [LRT]) is outside CarBlockArea.

already leaving the level crossing, the AV can indeed enter the level crossing. If the LRT is approaching the CarBlockArea, on the other hand, another principle has to be used. To this end, two operational conditions can be asserted.

$$T_{AV} = T_{\tau AV} + \frac{DS_{AV}}{V_{averageAV}}, T_{SignalLight} = \frac{Y}{V_{LRT}},$$
$$DS_{AV} = \begin{cases} \frac{V_{AVLimit}^2 - V_{av}^2}{2a}, & V_{AV} \geqslant V_{AVLimit} \\ \frac{-V_{AVLimit}^2 + V_{AV}^2}{2b}, & V_{AV} \leqslant V_{AVLimit} \end{cases} \quad (3)$$

A. If the time required by the AV to reach the CarSpeedLimit ($T_{AV}$) is greater than the time required for the LRT to reach the light signal ($T_{SignalLight}$), then the AV has to stop at the CarSpeedLimit sign. The respective times are calculated based on Equation 3, where $DS_{AV}$ is the acceleration or deceleration distance of the AV, $V_{averageAV}$ is the AV's average speed, $a$ is the acceleration of the AV, and $V_{AVLimit}$ is the AV speed limit. A principle similar to that in Figure 1 will be adopted. The AV will stop at the CarSpeedLimit for the time required by the LRT to complete the remaining journey to clear the level crossing.

$$S_{AV} = \begin{cases} V_{AV}T_{\tau AV} + \frac{V_{AVLimit}^2 - V_{AV}^2}{2a} + L + T_{dW} + D + T_{safe}, & V_{AV} \geqslant V_{AVLimit} \\ V_{AV}T_{\tau AV} + \frac{-V_{AVLimit}^2 + V_{AV}^2}{2b} + L + T_{dW} + D + T_{safe}, & V_{AV} \leqslant V_{AVLimit} \end{cases} \quad (4)$$

B. If the time required by the AV to reach the CarSpeedLimit is less than the time required by the LRT to reach the CarBlockArea, then the AV can enter the level crossing. Here, the possibility of a collision between AV and LRT may occur. Therefore, the LRT driver should drive cautiously, where $T_{dw}$ is the width of the level crossing for the track side, $T_{safe}$ is the LRT safe distance, and $D$ is the distance between the CarSpeedLimit sign and the crossing edge.

$$S_{LRT} = T_{rCl} + R_{dw} + T_L + T_rS_d + Y \quad (5)$$

If the LRT approaches while the AV is already inside the level crossing, the LRT driver will follow special procedures to avoid a collision. As a result, the total distance for the LRT to clear the level crossing can be obtained from Figure 2 as Equation 5, where $T_rS_d$ is the total distance of the CarBlockArea when the LRT reaches a light signal.

## *Concept of Optimization-Based AV Driving Strategy Model*

This study demonstrates the optimum AV driving strategy and mathematical models that describe how the AV crosses the level crossing without stopping at the CarSpeedLimit sign when the LRT enters the CarBlockArea. It is assumed that the AV and LRT are fully connected from the beginning, the rail and road tracks are located perpendicularly, and the weather conditions are good (Figures 1 and 2). Thus, the following driving strategies have been considered to avoid traffic accidents and reduce traveling time while crossing. (i) The AV finishes the journey (crosses the level crossing) before the LRT reaches the crossing. This kind of driving happens when the AV can complete the whole journey (clearing the level crossing) before the LRT reaches the crossing. The mandatory condition for such driving is that the LRT's traveling time to reach the crossing is greater than the time required by the AV to traverse the whole journey of clearing the level crossing. We could then look for the maximum time traveled by the AV

without causing a collision while clearing the level crossing. (ii) The LRT completes the journey before the AV reaches the crossing. In such driving conditions, the time required by the AV is greater than the time needed by the LRT to complete the crossing. (iii) The third driving condition is that the AV should stop at the CarSpeedLimit sign because collision is unavoidable no matter what AV driving strategy is considered. Moreover, the optimization principle can be used to ensure the three driving strategies, where the objective function is the time difference between the AV and the LRT. The major constraints for collision avoidance are related to the times required by the LRT and the AV, respectively.

*Proposed Modeling.* This study proposes an optimization strategy to enhance the performance of AVs at level crossings in the presence of LRT. Thus, a constrained objective function is formulated. It should be recalled that for the AV to cross the level crossing, the mandatory condition is that the time for the AV to clear the level crossing should be less than the time for the LRT to reach the crossing. Maximizing the time difference between the LRT and the AV is therefore considered the objective function. Additionally, the optimization gives the AV sufficient time to complete its journey before the LRT arrives at the level crossing.

$$J = \max\left(T_{LRT} - T_{AV}\right) \tag{6}$$

From the maximum value of the objective function, the chance of the AV clearing the level crossing can be deduced: meaning that, if $J$ is positive, then the AV can clear the level crossing without collision, and it should do so without stopping at the level crossing. Contrary to this, negative $J$ signifies that the AV cannot cross the level crossing without collision. Stoppage of the AV at the CarSpeedLimit sign will be enforced until the LRT clears the level crossing.

Safety measurement techniques are used to avoid unsafe circumstances at the level crossing region. These unsafe states are used as a constraint for the proposed optimization-based driving strategy. Thus, the constraints are as follows. (i) The AV's speed cannot be greater than its design speed. (ii) The AV's speed cannot be greater than the AV level crossing speed limit. (iii) The LRT's speed cannot be greater than the LRT level crossing speed. (iv) The LRT's speed cannot be greater than its design speed. (v) The AV cannot stop at the level crossing. The overall optimization problem becomes:

$$\begin{aligned} &J = \max\left(T_{LRT} - T_{AV}\right) \\ &\text{subjected to} \\ &\begin{cases} V_{LRT} \leqslant V_{LRTLimit}, V_{LRT} \leqslant V_{LRTDn} \\ V_{AV} \leqslant V_{AVLimit}, V_{AV} \leqslant V_{AVDn} \end{cases} \end{aligned} \tag{7}$$

where $V_{LRTLimit}$ is the limiting speed for the LRT to complete the journey from its current location to the level crossing, $V_{LRTLDn}$ is the LRT design speed, $V_{AVDn}$ is the AV design speed, and $T_{LRT}$ is obtained.

$$T_{LRT} = \frac{S_{LRT}}{V_{LRT}} \tag{8}$$

$$T_{AV} = T_{\tau AV} + T_{Dn} + T_{cs} \tag{9}$$

$$S_{Dn} = \begin{cases} \frac{V_{AVLimit} - V_{AV}}{a}, V_{AV} \geqslant V_{AVLimit} \\ \frac{-V_{AVLimit} + V_{AV}}{b}, V_{AV} \leqslant V_{AVLimit} \end{cases} \quad and \; T_{cs} = \frac{D + W + L + T_{safe}}{V_{AVLimit}} \tag{10}$$

Likewise, the AV's travel time can be calculated using Equation 9, where $T_{Dn}$ is acceleration/deceleration time until reaching the CarSpeedLimit sign, and $T_{cs}$ is the time required to clear the remaining level crossing distance at a constant speed of the level crossing speed limit. Thus, $S_{Dn}$ is the AV's acceleration/deceleration distance until reaching the CarSpeedLimit sign, and $S_{cs}$ is the clearance distance of the LRT. Consequently, these are obtained from Equation 10, where $W$ is track width.

## Model Simulation and Analysis

A MATLAB simulation was used to evaluate the performance of the proposed approach compared with the conventional stop-and-go driving strategy. The pseudo-code for the conventional stop-and-go approach and the proposed optimization-based driving methods are shown in Algorithms 1 and 2.

### *Conventional Driving Strategy Assignment Scheme*

As presented in the introductory section and the literature review, the existing conventional stopping driving strategy might be a significant constraint for mixed traffic safety. A summary of how to simulate the conventional driving strategy is given as Algorithm 1.

### *Optimization Driving Strategy Assignment Scheme*

This subsection describes the solving principle of the proposed optimization-based AV driving strategy. Thus, the constrained optimization AV driving model in the study must be converted to an unconstrained optimization problem. In this regard, a penalty approach is adopted, meaning the objective function will be penalized if the given state violates the constraints. The positive cost function $J$ indicates that the vehicle can cross the level crossing without collision. At the same time, the negative signifies that collision is unavoidable, and the stoppage of the AV will be enforced. Obtaining the best clearance time requires maximization of the cost function. It could

**Algorithm 1: Stop-and-go AV driving strategy**

```
Input: S_LRT, V_AV, V_AVLimit, a, b, D, W, L, T_τAV, T_L, L, T_Safe, R_dw, T_dw;
Output: T_AV, S_AV, T_LRT;
LRT_ClearDistance = S_LRT + T_L + T_dw + T_Safe
CarBlockArea = T_rS_d + T_L + T_dw + T_Safe
If (LRT_ClearDistance > CarBlockArea)
    Display ("LRT Approaching the CarBlockArea")
    T_SignalLight ← (S_T − T_rS_d)/V_LRT;
    DSAV ← AV distance to the CarSpeedLimit;
    T_AV ← DS_AV/V_averageAV + T_τAV;
    If (T_SignalLight < T_AV)
    Display ("Car should be stopped, CarSpeedLimit Signal Red");
  else
    Display ("Car can enter the level-crossing")
    S_AV ← Equation 4
    T_AV ← Calculate from S_AV
  end
else
  Display ("LRT is inside the CarBlockArea, and AV stops at the
    CarSpeedLimit")
  S_AV ← Equation 1
  T_AV ← Calculate from S_AV
end
```

be understood that the maximum value of the cost function ($J$) ascertains a good clearance time for the AV. Consequently, the optimization-based approach assures a safe clearance of the level crossing without collision. A summary of how to solve the proposed optimization-based AV driving pseudo-code is given as Algorithm 2. Since the study model adopted maximization, the penalty should be largely negative.

## Analysis of Existing Stop-and-Go Driving Approach

The simulation result indicates that when the LRT is 100 m inside the CarBlockArea, the LRT speed is 28.8 km/h, while the AV initial speed is 36 km/h, as shown in Figure 3. The AV level crossing speed limit is around 30 km/h. It can be easily observed that the AV stops at the CarSpeedLimit sign ($S_{AV}$ = D) (see Figure 3*a*) at 2.57 s. Since the LRT is inside the blocking region, the AV is forced to be in its stop state for the next LRT traversing period (14.42 s). In this range, under the stop-and-go principle, the AV is forced to stop even if it may possibly finish its journey without collision. The LRT continues its journey for 14.42 s after the AV stoppage time. At 16.98 s, the LRT has already cleared the level crossing. Therefore, after the LRT completes its journey, the AV can be freed using the signal light placed at CarSpeedLimit to continue its journey (Figure 3*b*).

The LRT approaching the CarBlockArea, and 1 m (a tight situation) away from the CarBlockArea, is shown in Figure 4. The AV is driven at 36 km/h while the LRT travels at 28.8 km/h. The AV level crossing speed limit is around 30 km/h. Therefore, the time required for the AV to reach the CarSpeedLimit sign is about 0.44 s, while the

**Algorithm 2: Optimization-based AV driving strategy**

```
Input: S_LRT, V_AV, V_AVLimitt, a, b, D, W, L, T_τAV, T_L, L, T_Safe, R_dw, T_dw Penalty = −10,000;
Output: T_AV, S_AV, T_LRT, J;
S_T = [0: S_LRT];
V_AVLimit = [0: V_AVLimit];
OptimumJ memory;
for j = 1: length(S_LRT)
    for i = 1: length(V_AVLimit)
        T_LRT ← Equation 9;
        TAV ← Equation 10;
        OptimumJ(j, i) ← T_LRT T_AV;
        if (violation)
        OptimumJ(j, i) = OptimumJ(j, i) + penalty;
    end
    Contour plot of ptimum;
    Indexj = Search S_T index from OptimumJ;
    [J, index] = max(OptimumJ(Indexj, :);
    if (J > 0)
        Display ("Car can clear the level-crossing without collision");
        CarSpeedatSignal = V_AVLimit(index);
        T_AV = [0, T_τAV, T_τAV + T_Dn, T_τAV + T_Dn + T_cs];
        DS_AV ← AV distance to the CarSpeedLimit;
        S_AV = [0, T_τAV*V_AV, T_τAV*V_AV + DS_AV, T_τAV*V_AV + DS_AV + T_cs*CarSpeedatSignal];
    else
        Display ("Car cannot clear the level-crossing without collision")
    end
end
```

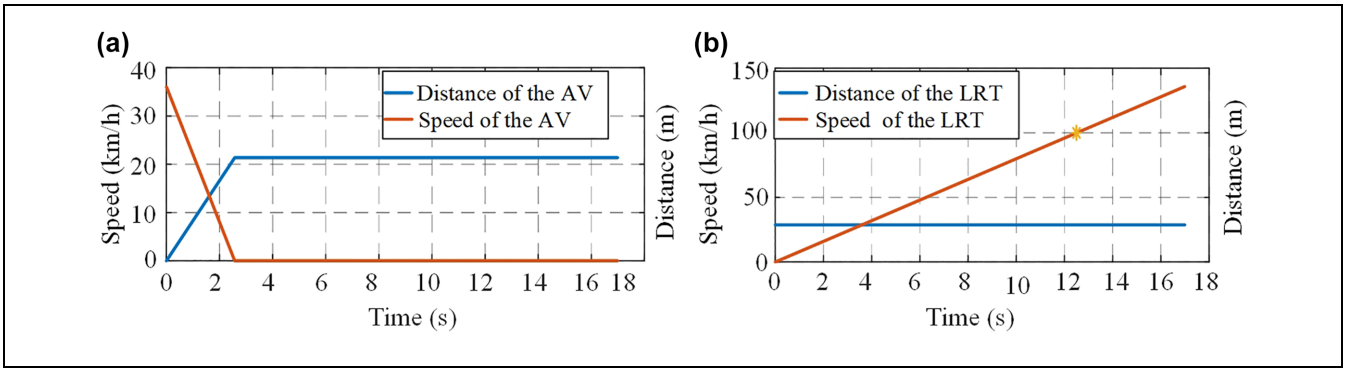



**Figure 3.** Simulation result for conventional stop-and-go driving when the train (light rail transit [LRT]) is inside the CarBlockArea: (*a*) autonomous vehicle driving and (*b*) LRT driving.

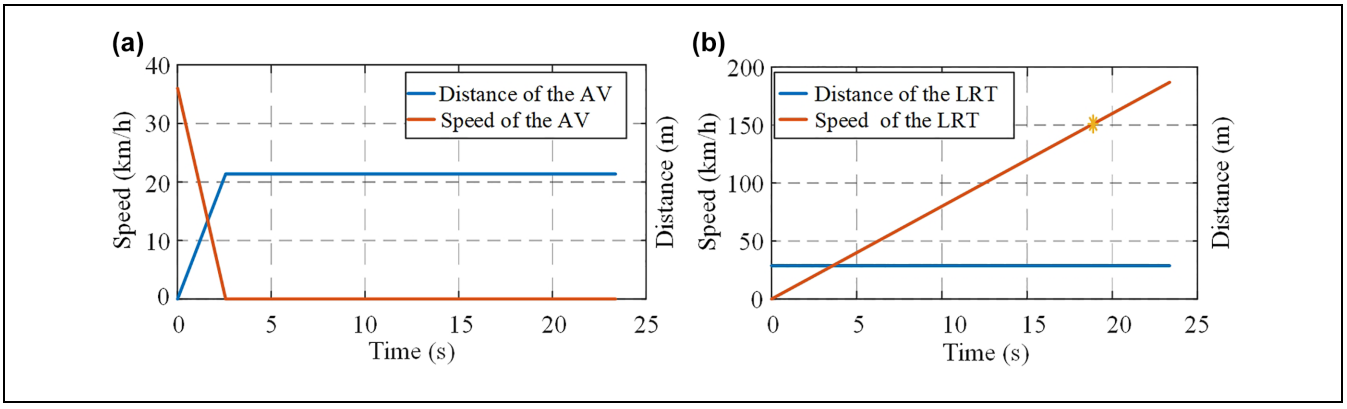



**Figure 4.** Simulation result for conventional stop-and-go driving when the initial speed of the autonomous vehicle (AV) is 36 km/h and the train (light rail transit [LRT]) is 1 m from the CarBlockArea at a speed of 28.8 km/h: (*a*) AV driving and (*b*) LRT driving.

time required by the LRT to reach the CarBlockArea is 0.13 s. This shows that the LRT reaches the CarBlockArea before the AV reaches the CarSpeedLimit. Consequently, the AV cannot enter the level crossing under the stop-and-go principle. Therefore, the AV decelerates from 36 km/h to zero and stops until the LRT clears the level crossing (for about 20.8 s) (Figure 4*a*). Figure 4*b* shows the LRT driving principle. Finally, at 23.4 s, the LRT clears the level crossing and the CarSpeedLimit will be cleared to allow the AV to cross.

Likewise, Figure 5 indicates an operation for the LRT far away from the CarBlockArea. The LRT is located 50 m away from the CarBlockArea (or light signal) with a speed of 28.8 km/h. The AV's initial speed is 36 km/h. The AV level crossing speed limit is around 30 km/h. The time required by the AV to reach the CarSpeedLimit value (30 km/h) is 0.44 s, while the LRT needs around 6.25 s to enforce AV blockage. Thus, the AV decelerates to the level crossing speed limit and traverses the level crossing. It can be clearly shown in Figure 5*a* that the time required by the AV to clear the level crossing is 1.67 s, while the time required by the LRT to reach the level crossing is around 25 s. Therefore, no collision will happen. However, the LRT driver could drive cautiously to avoid unnecessary scenarios that may lead to a collision. Therefore, a cautious LRT driving strategy avails in Figure 5.

If the AV level crossing speed limit is 1.44 km/h and the AV initial speed is 36 km/h while the LRT is 50 m away from the CarBlockArea at a speed of 28.8 km/h, then a special driving strategy is availed of. The simulation result for this state is shown in Figure 6. Likewise, in Figure 5, the time required by the AV to reach the CarSpeedLimit is 2.46 s, while the LRT needs around 6.25 s to enforce AV blockage. Consequently, the AV is free to enter the level crossing. However, the time required by the AV to clear the level crossing is 27.1 s, and the LRT requires 25 s to reach the level crossing. This indicates that there is a high probability of collision if the LRT is traveling at a speed of 28.8 km/h. Thus, a collision will happen unless the LRT driver is driving cautiously. This shows that if the CarSpeedLimit value is below a certain point, the possibility of collision is high and traffic safety is reduced. Moreover, the LRT driver

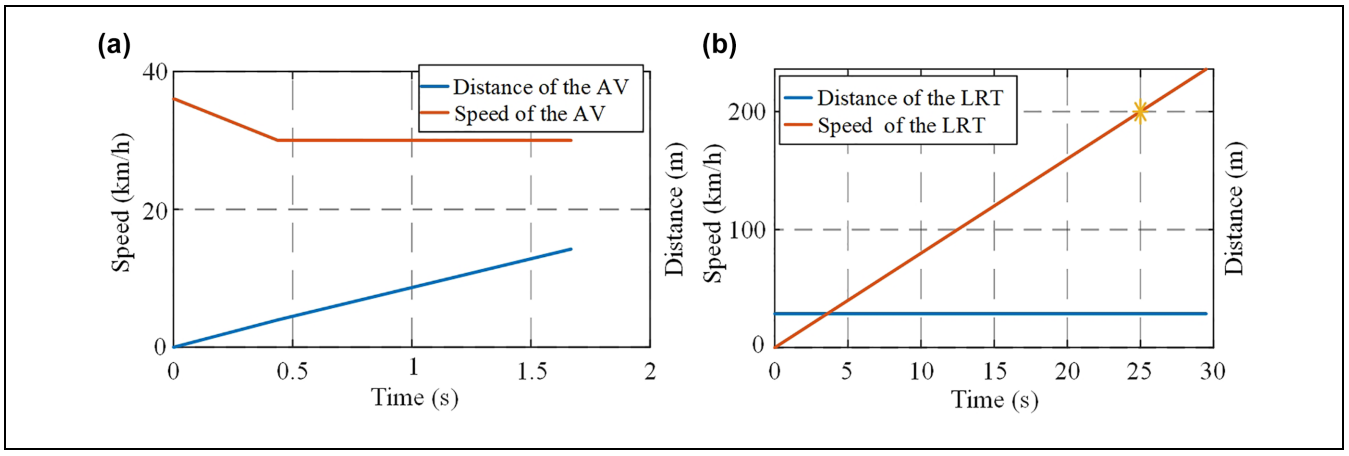



**Figure 5.** Simulation result for conventional stop-and-go driving when the initial speed of the autonomous vehicle (AV) is 36 km/h and the train (light rail transit [LRT]) is 50 m from the CarBlockArea at a speed of 28.8 km/h: (*a*) AV driving and (*b*) LRT driving.

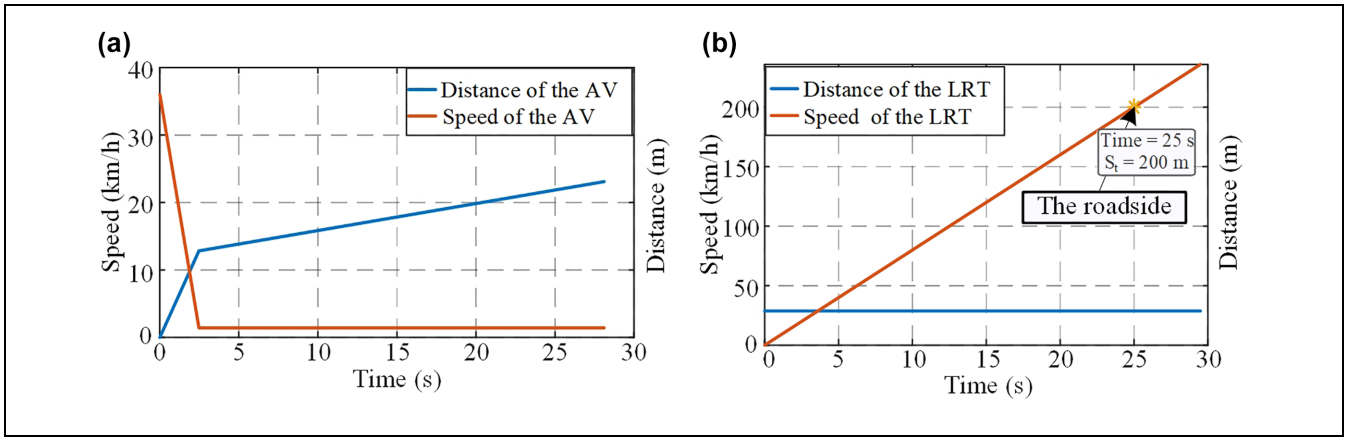



**Figure 6.** Simulation result for conventional stop-and-go driving when the initial speed of the autonomous vehicle (AV) is 36 km/h and the train (light rail transit [LRT]) is 50 m from the CarBlockArea at a speed of 28.8 km/h: (*a*) AV driving and (*b*) LRT driving.

could consider the necessary attention if the traffic signal releases the vehicle before the LRT reaches the CarBlockArea.

The simulation results showed that the stop-and-go driving strategy will always enforce the AV stopping if the LRT enters the CarBlockArea. Therefore, although traffic safety can be ensured, such a driving strategy is susceptible to traffic loss. Besides, suppose the LRT is outside the CarBlockArea, and the AV level crossing speed limit is below a certain value. In that case, the probability of collision occurrence increases and adds an unnecessary burden on the LRT driver to avoid such inconvenience.

## *Analysis of Optimization-Based Driving*

Figure 7 shows the operation performance of the proposed method for LRT 1 m away from the CarBlockArea with the LRT speed of 28.8 km/h, while the AV initial speed is 36 km/h. The AV is approaching the CarSpeedLimit with speed greater than the maximum speed limit of the CarSpeedLimit. Figure 7*c* is the $J$ values for the CarSpeedLimit ranging from zero to the maximum speed limit. It can be observed that $J$ becomes negative for a speed limit less than 0.62 km/h. This signifies that the AV can clear the level crossing for a speed above 0.62 km/h without causing a collision at the level crossing. If by any means, the level crossing speed limit is less than 0.62 km/h, then the point of collision is unavoidable. The conventional stop-and-go method should be adopted here, meaning that the AV has to stop at the CarSpeedLimit until the LRT clears the level crossing. The maximum $J$ is obtained at the 28.8 km/h speed limit and the AV can clear the level crossing without collision. The speed profile to be followed by the AV is given in Figure 7*a*. First, the AV has to decelerate from the

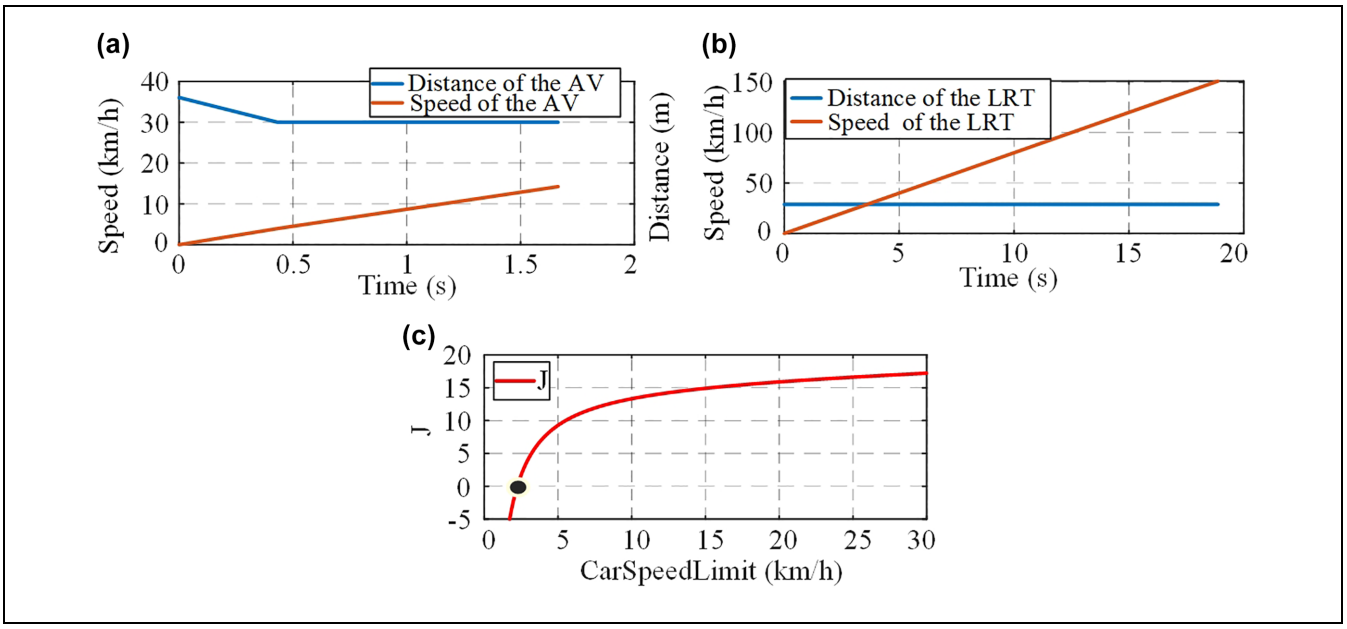



**Figure 7.** The optimum-based approach for the autonomous vehicle (AV) driving with an initial speed of 36 km/h and the train (light rail transit [LRT]) at 1 m from the CarBlockArea at 28.8 km/h, (*a*) AV driving, (*b*) LRT driving, and (*c*) AV optimum speed limit.

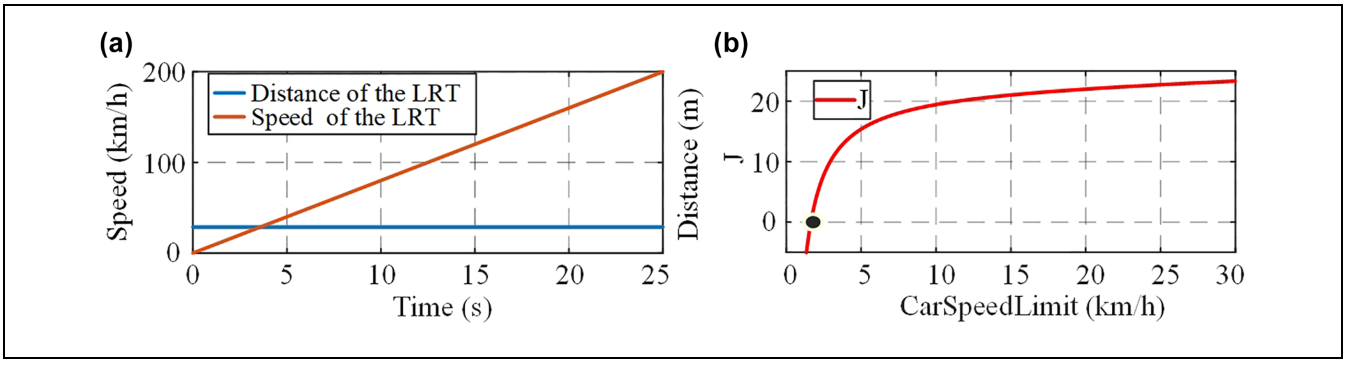



**Figure 8.** Optimum performance of autonomous vehicle (AV) driving at an initial speed of 36 km/h and train (light rail transit [LRT]) 50 m from the CarBlockArea at 28.8 km/h, (*a*) LRT driving and (*b*) AV optimum speed limit.

current speed (36 km/h) to the vehicle signal speed limit of the level crossing (28.8 km/h), and then with the constant speed, the AV will clear the level crossing at 1.66 s.

The total distance traversed by the AV is 14.23 m. The LRT speed profile is shown in Figure 7*b*. Compared with the stop-and-go driving with a similar LRT location shown in Figure 4*a*, this study approach avoids AV stoppage. The AV can only be cleared after 23.4 s under the stop-and-go strategy, while the AV can complete the whole journey within 1.66 s using the proposed optimization-based approach. The proposed optimization-based AV driving strategy thus saves traffic volume, avoids delays in traveling time, and ensures traffic safety.

The optimization-based AV driving strategy when the LRT is 50 m away from the CarBlockArea is shown in Figure 8. The LRT speed is 28.8 km/h, while the AV initial speed is 36 km/h. It can be understood from the optimization value *(J)* in Figure 8*b* that if the level crossing speed limit is less than 1.64 km/h, the AV cannot be allowed to enter the level crossing. The maximum *J* is obtained for the level crossing speed limit of 28.8 km/h. The simulation result for AV driving is therefore as in Figure 7*a*. The LRT driving strategy is shown in Figure 8*a*. One can observe that the AV driving strategy of the proposed method and the stop-and-go approach shown in Figure 5*a* are similar. However, the optimum *J* value

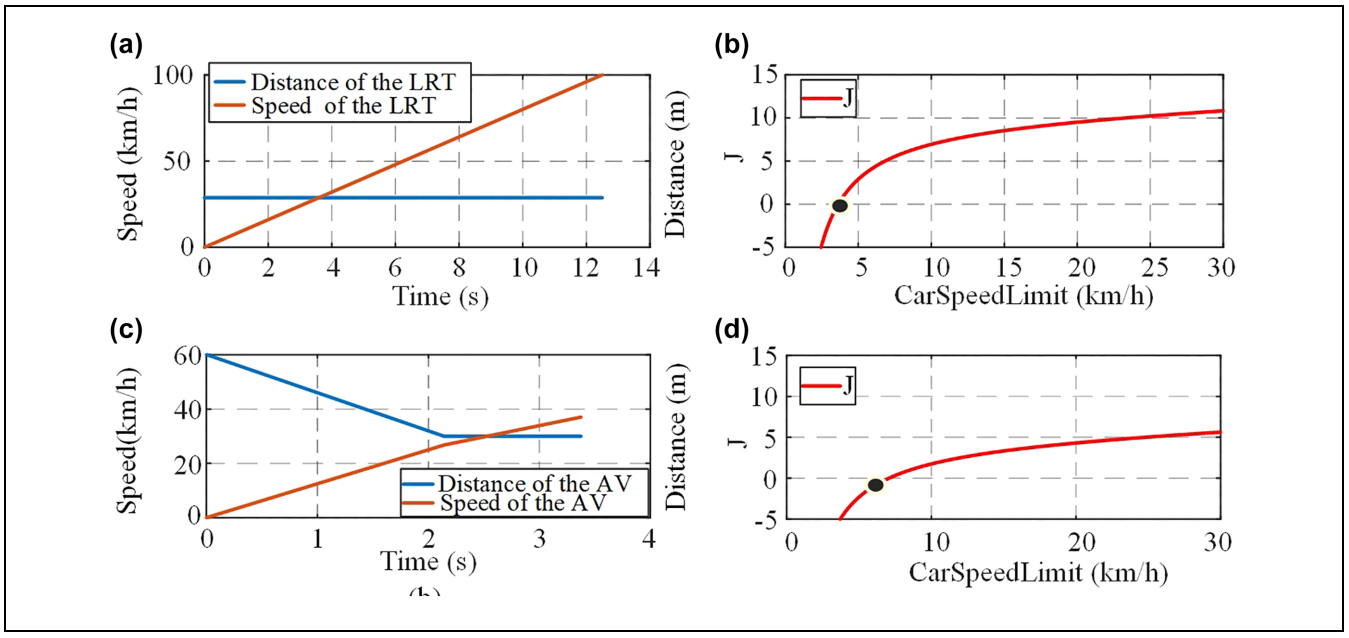



**Figure 9.** Optimum performance of autonomous vehicle (AV) driving at an initial speed of 36 km/h, 60 km/h, and light rail transit (LRT) at 100 m inside CarBlockArea within 28.8 km/h: (*a*) LRT driving, (*b*) AV optimum speed limit, (*c*) AV driving, and (*d*) AV optimum speed limit.

as shown in Figure 8*b* indicates that if the speed limit is less than 1.64 km/h, then the AV should stop at the CarSpeedLimit to avoid a collision. Compared with Figure 6, showing the stop-and-go strategy, the proposed method shows that the AV can enter the level crossing area as the signal releases, leading to a safety constraint. Thus, this optimization-based approach can be called the collision avoidance method. The LRT driver has no extra burden as the system assures traffic safety.

Figure 9 shows the simulation result where the LRT is inside the CarBlockArea. The LRT and AV initial speeds are 28.8 km/h and 36 km/h, respectively, as depicted in Figure 9, *a* and *c*. The initial speed of the AV is 60 km/h, as shown in Figure 9, *b* and *d*. The LRT is located 100 m from the roadside. The optimum value of $J$ is shown in Figure 9, *c* and *d*. The AV will be driving at a speed profile as shown in Figures 7*a* and 9, *b* and *c*, as the maximum $J$ is at the CarSpeedLimit of 28.8 km/h. The negative $J$ value conveys that if the CarSpeedLimit at the level crossing is approximately below 3.64 km/h and 4.34 km/h, respectively, the AV cannot enter the level crossing. Therefore, the AV will stop at the CarSpeedLimit until the LRT clears the level crossing. Compared with the stop-and-go approach (see Figure 3*a*), this optimization-based approach allows the AV to enter the level crossing, although the LRT is inside the CarBlockArea. Therefore, this study approach improves traffic safety.

The contour plot of $J$ in relation to LRT location and CarSpeedLimit is shown in Figure 10. It shows the state at which the AV cannot enter the level crossing in relation to LRT location and CarSpeedLimit. Negative $J$ signifies that a collision may occur at the level crossing, and the conventional stop-and-go principle must be in place. On the other hand, the positive $J$ indicates that the AV can cross the level crossing without safety concerns. The shaded area indicates the AV cannot enter the level crossing. Figure 10*a* shows the AV initial speed at 36 km/h, Figure 10*b* shows the contour plot of $J$ for the AV initial speed of 10 km/h. As the speed limit decreases, the area for which the AV can be allowed to enter the level crossing decreases. Thus, the proposed approach decides whether the AV can enter the level crossing based on the AV's initial speed, the AV level crossing speed limit, and the location of the LRT.

## Comparison of the Study

Comparing the conventional stop-and-go and the proposed driving approaches convinces us that the proposed method ensures traffic safety. The LRT speed at the level crossing is assumed to be 28.8 km/h and the AV's initial speed 36 km/h. Other AV initial speed values can also be taken, and the respective results can be easily obtained. If the LRT is outside of the CarBlockArea, the stop-and-go approach will clear the AV to enter the level crossing based on the time required to reach the CarSpeedLimit sign and the CarBlockArea for the AV and the LRT, respectively. If the CarSpeedLimit signal is clear, the LRT can enter the level crossing. However, the time required by the AV to clear the level crossing might be greater than the time required by the LRT to reach the level crossing leading to a high probability of collision.

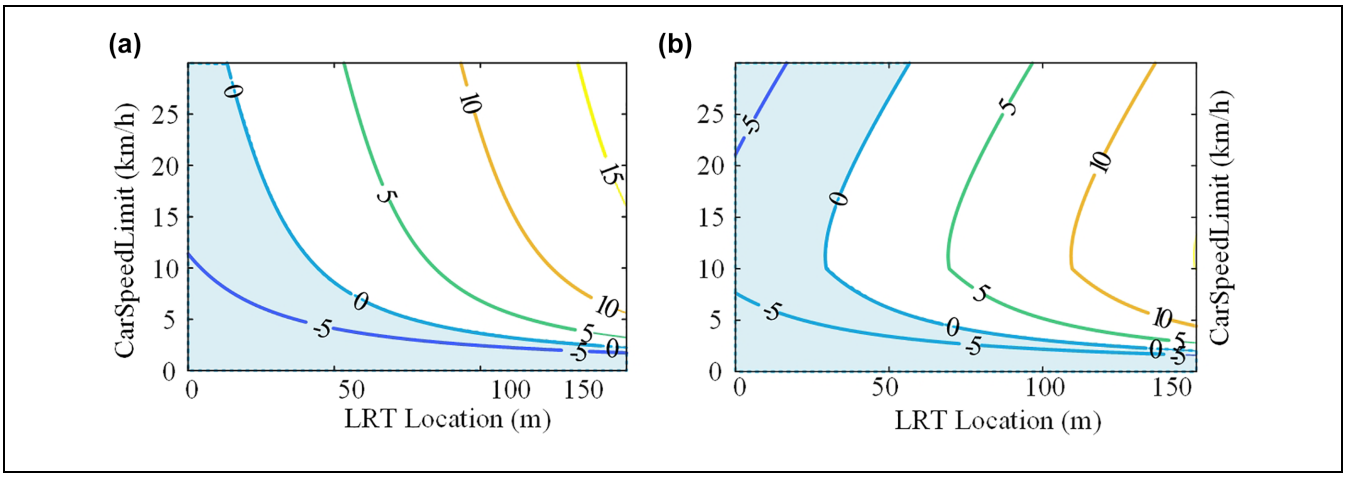



**Figure 10.** Contour plot of *J* with initial speed of autonomous vehicle (AV) at (*a*) 36 km/h and (*b*) 10 km/h.

Such a driving condition is shown in Figure 6. Although the CarSpeedLimit signal clears the AV, the time required by the AV to clear is 27.1 s, and the LRT requires 25 s to reach the roadside. Thus, the LRT driver will be forced to drive cautiously and an extra burden will be added besides traffic safety constraints. The proposed approach, however, can avoid such driving constraints. The AV can only enter the level crossing based on the time required by the AV and the LRT. Figure 8 shows an optimization-based driving approach. The AV can only enter the level crossing if the time required by the AV to clear the level crossing is less than the time taken by the LRT to reach the crossing.

The *J* values signify whether the AV enters the level crossing or not. Thus, if the CarSpeedLimit is less than 1.64 km/h, the AV will not be allowed to enter the level crossing if its initial speed is 36 km/h. This ensures traffic safety. The one thing required for such driving is that the AV should continuously receive the LRT's location while the AV is approaching the level crossing. The stop-and-go method unilaterally halts the AV if the LRT is inside the CarBlockArea. Thus, there is a loss of traffic flow. On the other hand, the optimization-based approach depends on the location of the LRT. The optimization-based approach, with AV driving inside the CarBlockArea, is shown in Figure 9. It can be observed that *J* is positive for CarSpeedLimit greater than 3.64 km/h. This conveys that the LRT can enter the level crossing without collision and adds an extra constraint to the LRT driver. Therefore, the proposed approach saves traffic delay and ensures safety compared with the stop-and-go strategy. The contour plot of *J* for the LRT inside and outside the CarBlockArea while the AV CarSpeedLimit at the level crossing is shown in Figure 10, *a* and *b*, respectively. The shaded area signifies that the AV cannot enter the level crossing because of the safety issues (Figure 10).

A summary of the comparison between the proposed approach and the conventional stop-and-go approach for various AV initial speeds is shown in Table 1. Overall, the proposed method provides safety-assured AV driving based on LRT location, initial speed of the AV, and CarSpeedLimit of the level crossing.

## Conclusion

Level crossings present a serious threat to the safety and mobility of drivers near railroad crossings. They affect the flow of traffic and collisions occur. The existing conventional stop-and-go safety principle is not relevant for mixed traffic since it forces the AV to halt at CarSpeedLimit within a minimum of 4.5 m from the level crossing. Therefore, this study proposes an optimization-based AV driving strategy to improve the traffic safety of AVs at level crossings. An optimization model considering the operational constraints is developed. The cost function of the proposed optimization-based driving strategy is the time difference between the time required by the LRT to reach the crossing and the time required for the AV to complete the whole level crossing. Where the AV's speed cannot be greater than its design speed, the AV's speed cannot be greater than the AV level crossing speed limit, the LRT's speed cannot be greater than the LRT's level crossing speed, and cannot be greater than its design speed, the AV cannot stop at the level crossing and these are considered as constraints. A MATLAB simulation model is developed to validate the proposed driving strategy. The simulation results indicate that it can have a significant impact on the safety of mixed traffic to avoid traffic loss when the LRT travels at 28.8 km/h. More specifically, the study demonstrated that the maximum time (*J*) difference between the LRT and AV travel is possible without causing a collision while clearing the level crossing. Thus, a negative *J*

**Table 1.** Comparison of conventional and Optimum J driving performance strategy in different Train State Positions and Initial Speeds of Autonomous Vehicle (AV) .

| | | Train outside and inside of CarBlockArea | | |
|---|---|---|---|---|
| | AV initial speed | 1 m away | 50 m away | 100 m inside the CarBlockArea |
| Conventional driving strategy | 10 km/h<br>36 km/h<br>60 km/h | Cannot enter the level crossing even for 30 km/h CarSpeedLimit | Can enter the level-crossing, but the train driver must drive cautiously | The AV cannot enter the level-crossing area |
| Optimization-based driving | 10 km/h | Enter the level crossing for CarSpeedLimit > 2.03 km/h | Enter the level crossing for CarSpeedLimit > 1.53 km/h | Enter the level crossing for CarSpeedLimit > 3.1 km/h |
| | 36 km/h | Enter the level crossing for CarSpeedLimit > 2.25 km/h | Enter the level crossing for CarSpeedLimit > 1.64 km/h | Enter the level crossing for CarSpeedLimit > 3.64 km/h |
| | 60 km/h | Enter the level crossing for CarSpeedLimit > 2.5 km/h | Enter the level crossing for CarSpeedLimit > 1.8 km/h | Enter the level crossing for CarSpeedLimit > 4.3 km/h |

indicates that the AV cannot enter the crossing level and vice versa. Consequently, compared with the stop-and-go driving principle with the LRT in a similar location, the proposed strategy significantly avoids AV stoppage at CarSpeedLimit. For instance, if the LRT is inside the CarBlockArea, the conventional stop-and-go approach immediately asserts the AV stoppage. Besides, if the LRT is 1 m away from the light signal (a tight situation), the AV will halt at CarSpeedLimit until the LRT clears the level crossing. On the other hand, the proposed method asserts the AV stoppage below a certain speed limit at the level crossing while considering the AV's initial speed. For instance, if the LRT is 100 m away from the level crossing, the proposed method asserts AV stoppage for speed limit less than 3.1 km/h, 3.64 km/h, and 4.3 km/h for initial speed of the AV at 10 km/h, 36 km/h and 60 km/h, respectively. Therefore, this optimization-based AV driving strategy has a good impact on improving traffic safety. Moreover, this strategy is appropriate for mixed traffic use as a safety assurance method. Therefore, the proposed optimization-based approach can be called a collision avoidance driving strategy. The LRT driver has no extra burden as the system assures traffic safety and saves traffic loss and traveling time.

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## Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: B.Y. Adane, B. Peng, J. Hou; data collection: B.Y. Adane; analysis and interpretation of results: B.Y. Adane, B. Peng, J. Hou, D.Y. Abate; draft manuscript preparation: B.Y. Adane, B. Peng, J. Hou, D. Y. Abate. All authors reviewed the results and approved the final version of the manuscript.

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